

Model of Intergroup Conflict — components and full mathematical formulation

Below is a single, self-contained description you can copy and share. It starts with the conceptual components (so the listener understands the pieces) and finishes with the compact mathematical model (the equations and how to use them). Tone is practical and precise — explainable to social scientists, programmers, or creative collaborators.

1) Purpose and scope

A dynamical model that explains how individual psychological tendencies, subgroup roles, triggers, narratives, power actors, social networks, and structural asymmetries combine to produce, amplify, or dampen intergroup (race) conflict.

It is mesoscopic: not an agent-by-agent simulation by default, but a coupled dynamical system of subgroup activations plus environmental modulators and feedback. Suitable for conceptual explanation, theoretical analysis (stability/tipping points), and straightforward numerical simulation.

2) Core conceptual components (plain language)

A. Psychological basis (individual knobs)

Each person has tunable internal axes that respond to events:

- **Threat / fear sensitivity** — how readily one perceives danger.
- **Identity salience** — how strongly one thinks in group terms.
- **Anomaly bias** — tendency to mark unfamiliar as threatening.
- **Reciprocity / moral punishment drive** — tendency to generalize retaliation.
- **Empathy / cognitive complexity** — capacity to individuate or contextualize.
- **Historical memory** — weight of past grievances.

These are the “hardware” on which social dynamics run.

B. Subgroup roles (aggregates)

Within each population we track three (minimal) aggregate states:

- **T(t) — Triggerers**: a small, loud subgroup that initiates conflict (high threat, high identity, low cross-group empathy).
- **V(t) — Vulnerable / Mobilizable**: the swing majority/minority that is pulled into conflict under fear or social pressure.
- **S(t) — Resistive / Moderates**: individuals who dampen conflict (higher empathy, lower threat, stronger norms of individuated judgment).

Each variable represents the *activation level* or effective proportion of the population in that role (scaled 0–1 or as continuous activations).

C. Triggers (forcings)

Events that push activations:

- **N(t)** — natural triggers (accidents, resource shocks, random violence).
- **G_{in}(t)** — ingroup-created triggers (propaganda, rituals, leaders amplifying fear).
- **G_{out}(t)** — outgroup-created triggers (provocations — intentional or interpreted as such).

D. Power, narratives, institutions

- **P(t)** — influence of power-seekers who intentionally manipulate fear/identity to gain or keep control.
- **C(t)** — cultural/narrative shaping (media, education, myths) that slowly tune baseline parameters.
- **Inst(t)** — institutional capacity / trust that provides negative feedback (law, fair enforcement, mediation).

E. Structural / network modulators

- **R_e(t)** — structural asymmetry (economic/political power differentials, demographic gradients).
- **L(t)** — network connectivity / exposure (how quickly activation spreads across people and groups).

These scale sensitivities and rates of spread.

F. Feedback and time-scales

- **Positive feedback:** conflict increases future triggers (cycles of retaliation, rumor, and political exploitation).
- **Negative feedback:** strong institutions, resistive group action, or cross-group contact dampen conflict.
- **Multiple time-scales:** fear spikes quickly; identity and narratives change slowly; institutions and structures evolve on long time-scales.

3) Notation and variable summary (for the equations)

- **T(t)** — activation of triggerers.
- **V(t)** — activation of vulnerable group.
- **S(t)** — activation of resistive/moderate group.
- **R(t)** — conflict intensity (output).

- $N(t)$ — natural triggers (external).
- $G_{in}(t), G_{out}(t)$ — ingroup/outgroup created triggers.
- $P(t)$ — power-seeker influence.
- $C(t)$ — cultural / narrative shaping.
- $Inst(t)$ — institutional trust / enforcement strength.
- $R_e(t)$ — structural asymmetry index.
- $L(t)$ — network connectivity coefficient.
- $\epsilon_{\{\cdot\}}(t)$ — stochastic noise terms (optional).

All variables are functions of time t . Coefficients (Greek letters below) are positive constants to be estimated or chosen for simulations.

4) Compact mathematical model (coupled dynamical system)

4.1 Subgroup activation dynamics

Here we write conceptual differential equations for how subgroup activations change over time.

Triggerers:

$$\frac{dT}{dt} = \alpha_N N(t) + \alpha_{in} G_{in}(t) + \alpha_{out} G_{out}(t) + \alpha_P P(t) + \alpha_V V(t) - \alpha_S S(t) - \kappa_T T(t) + \epsilon_T(t)$$

Vulnerable group:

$$\frac{dV}{dt} = \beta_T T(t) \cdot L(t) + \beta_N N(t) + \beta_P P(t) \cdot (1 + \phi R_e(t)) - \beta_S S(t) - \kappa_V V(t) + \epsilon_V(t)$$

Resistive group:

$$\frac{dS}{dt} = \gamma_C C(t) + \gamma_{Inst} Inst(t) - \gamma_T T(t) - \gamma_P P(t) - \kappa_S S(t) + \epsilon_S(t)$$

Explanation of terms:

- Positive terms add to activation (triggers, contagion via L , power influence scaled by asymmetry R_e).
- Negative terms represent suppression (resistive group dampening), natural relaxation κ (decay toward baseline), and suppression by power/triggerers where relevant.
- $\epsilon_{\cdot}(t)$ are small noise processes (e.g., white noise or Poisson shock processes) to model chance events.

4.2 Conflict intensity (observable)

Conflict intensity $R(t)$ is an increasing function of triggerer and vulnerable activation, offset by resistive power:

A simple choice:

$$R(t) = \rho_T T(t) + \rho_V V(t) - \rho_S S(t)$$

with $R(t) \geq 0$ (apply $\max(0, \cdot)$ if needed). More nonlinear forms can be used (e.g., multiplicative effects or thresholding).

4.3 Trigger feedback dynamics

Past conflict increases future triggers (positive feedback). We capture this discretely or continuously. Continuous form (example):

$$\begin{aligned}\frac{dG_{in}}{dt} &= \mu_{in}R(t) - \kappa_{gin}G_{in}(t) \\ \frac{dG_{out}}{dt} &= \mu_{out}R(t) - \kappa_{gout}G_{out}(t) \\ \frac{dN}{dt} &= \mu_N R(t) - \kappa_N N(t) + \eta(t)\end{aligned}$$

Here μ . capture how much conflict generates further created triggers or natural cascade; $\eta(t)$ is exogenous shocks.

4.4 Slow cultural tuning and power shaping

Narratives and power reshape baseline sensitivities slowly (long time-scales). Represent as slow differential equations:

$$\begin{aligned}\frac{dC}{dt} &= \sigma_C P(t) - \kappa_C C(t) \\ \frac{dP}{dt} &= \sigma_P R(t) - \kappa_P P(t)\end{aligned}$$

Interpretation: power grows when conflict is useful; narratives are reinforced by power; both decay slowly if usefulness drops.

5) Structural modulations and network effects (how to plug them in)

- **Network connectivity** $L(t)$ multiplicatively scales contagion from T to V . High $L \rightarrow$ fast spread.
- **Structural asymmetry** $R_e(t)$ increases the effectiveness of power manipulation and the sensitivity of V to $P(t)$. For example, use factor $(1 + \phi R_e(t))$ on terms influenced by P .

- **Institutions** $Inst(t)$ directly increase S (resistive force) and can reduce propagation by lowering β_T or increasing K_V .

6) Practical implementation notes (no heavy math required)

- Choose state scaling: all activations scaled between 0 and 1, or use counts/proportions.
- Pick time units (days, weeks). Choose relaxation rates K consistent with time units (e.g., fear decays in days $\rightarrow$ large K ; cultural shifts decay in years $\rightarrow$ small K).
- Start with a minimal parameter set for experiments: set many small coefficients to 0 to see isolated effects, then enable coupling.
- Add noise terms $\epsilon(t)$ to model surprise events and test sensitivity.
- Explore two-group versions first (Group A, Group B), or collapse symmetrical assumptions for simplicity.

7) Analysis roadmap (how to study the model)

- **Equilibria & stability:** find steady states (low-conflict vs high-conflict) and analyze sensitivity to α, β, γ .
- **Bifurcation / tipping points:** vary P , R_e , or L to find thresholds where $R(t)$ jumps.
- **Network experiments:** vary L and clustering to see how echo chambers change speed and amplitude.
- **Policy tests:** increase $Inst$ or C (pro-social narratives) and measure reduction in peak R .
- **Agent-based extension:** if desired, convert individual-level parameters (threat, empathy, identity) into agents on a social graph and use same update rules to reproduce macro-level behaviors.

8) Assumptions and limitations (be explicit when communicating)

- The model abstracts complex individual psychology into a few continuous axes — it is a simplification.
- Parameter values and functional forms are choices; qualitative behavior (feedback, tipping, slow/fast dynamics) is robust, exact thresholds are model-dependent.
- Cultural meaning and historical contingencies can be richer than captured by a single $C(t)$ variable; treat that variable as a placeholder for many slow cultural processes.
- Ethical and empirical caution: use anonymized, aggregate data when calibrating; the model is for explanation, scenario testing, and policy insight — not for targeting individuals or groups.

9) Short textual summary you can paste in a meeting or paper

We model intergroup conflict as a coupled dynamical system of three aggregate subgroups — triggerers (T), vulnerable/mobilizable (V), and resistive/moderates (S) — driven by external triggers (natural and created), power and narrative influence, network contagion, and structural asymmetries. Subgroup activations follow simple differential

rules: triggers and power increase T and V (scaled by network connectivity and asymmetry), resistive forces and institutions suppress them. Conflict intensity $R(t)$ is a function of T and V minus S . Feedback loops let conflict generate more triggers and strengthen power-holders, while slow cultural dynamics tune baseline sensitivities. The model is designed to study thresholds, feedback, and policy levers (strengthening institutions or pro-social narratives) without requiring fine-grained agent data.

If you want, I can now:

- produce a **one-page PDF** of the above,
- or generate an **illustrative flow diagram** (boxes/arrows) mapping the equations to the conceptual pieces,
- or propose a minimal set of default parameter values and a short simulation script (python pseudocode) you can hand to a programmer.

Which of those would you like next?